Numerical Methods

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = x^3 - 4x - 2$ has a root between $x = 2$ and $x = 3$.

$f(2) = 8 - 8 - 2 = -2$. $f(3) = 27 - 12 - 2 = 13$. Since $f(2) < 0$ and $f(3) > 0$, the sign change confirms a root lies in $(2, 3)$.

2 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = x^3 - x - 1$ has a root between $x = 1$ and $x = 2$.

$f(1) = 1 - 1 - 1 = -1$. $f(2) = 8 - 2 - 1 = 5$. Since $f(1) < 0$ and $f(2) > 0$, the sign change confirms a root lies in $(1, 2)$.

3 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = \ln x - 2 + x$ has a root between $x = 1$ and $x = 2$.

$f(1) = \ln 1 - 2 + 1 = -1$. $f(2) = \ln 2 - 2 + 2 = \ln 2 \approx 0.693$. Since $f(1) < 0$ and $f(2) > 0$, the sign change confirms a root lies in $(1, 2)$.

4 [Foundational] • Target: Locating roots by a change of sign

Show that $f(x) = x^3 - 2x - 5$ has a root between $x = 2$ and $x = 3$.

$f(2) = 8 - 4 - 5 = -1$. $f(3) = 27 - 6 - 5 = 16$. Since $f(2) < 0$ and $f(3) > 0$, the sign change confirms a root lies in $(2, 3)$.

5 [Foundational] • Target: Interval bisection

Using $f(x) = x^3 - x - 1$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval containing the root.

$f(1.5) = 3.375 - 1.5 - 1 = 0.875 > 0$. Since $f(1) < 0$ and $f(1.5) > 0$, the root lies in $(1, 1.5)$.

6 [Foundational] • Target: Interval bisection

Continuing from the previous problem, evaluate $f(1.25)$ and give a narrower interval for the root.

$f(1.25) = 1.953125 - 1.25 - 1 = -0.296875 < 0$. Since $f(1.25) < 0$ and $f(1.5) > 0$, the root lies in $(1.25, 1.5)$.

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7 [Foundational] • Target: Interval bisection

Using $f(x) = \ln x - 2 + x$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval for the root.

$f(1.5) = \ln 1.5 - 2 + 1.5 = 0.405465 - 0.5 = -0.094535 < 0$. Since $f(1.5) < 0$ and $f(2) > 0$, the root lies in $(1.5, 2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Fixed-point iteration

The equation $x^3 - 4x - 2 = 0$ rearranges to $x = \sqrt[3]{4x + 2}$. Using $x_0 = 2$, find x_1, x_2, and x_3, to four decimal places.

$x_1 = \sqrt[3]{4(2) + 2} = \sqrt[3]{10} \approx 2.1544$. $x_2 = \sqrt[3]{4(2.1544) + 2} = \sqrt[3]{10.6176} \approx 2.1978$.
$x_3 = \sqrt[3]{4(2.1978) + 2} = \sqrt[3]{10.7912} \approx 2.2098$ (converging toward the root ≈ 2.214).

9 [Intermediate] • Target: Fixed-point iteration

The equation $x^2 - 5x + 2 = 0$ rearranges to $x = 5 - \frac{2}{x}$. Using $x_0 = 4$, find x_1 and x_2, to four decimal places.

$x_1 = 5 - \frac{2}{4} = 4.5$. $x_2 = 5 - \frac{2}{4.5} \approx 5 - 0.4444 = 4.5556$.

10 [Intermediate] • Target: Fixed-point iteration

The equation $x^3 - 2x - 5 = 0$ rearranges to $x = \sqrt[3]{2x + 5}$. Using $x_0 = 2$, find x_1, to four decimal places.

$x_1 = \sqrt[3]{2(2) + 5} = \sqrt[3]{9} \approx 2.0801$.

11 [Intermediate] • Target: Convergence and divergence of iterations

Explain why the iteration $x_{n+1} = x_n^3 - 3$ (i.e. $g(x) = x^3 - 3$), starting from $x_0 = 1.5$, would fail to converge to a root of $x^3 - x - 3 = 0$ near $x = 1.5$.

$g'(x) = 3x^2$. At $x = 1.5$: $g'(1.5) = 3(2.25) = 6.75$. Since $|g'(1.5)| = 6.75 > 1$, the iteration does not satisfy the convergence condition $|g'(x)| < 1$ near this point, so it diverges rather than converging to the root.

12 [Intermediate] • Target: Convergence and divergence of iterations

Using $g(x) = 5 - \frac{2}{x}$ from the earlier iteration, find $g'(x)$ and evaluate $|g'(x)|$ at $x = 4.56$ (near the root), to explain why the iteration converges there.

$g(x) = 5 - 2x^{-1}$, so $g'(x) = \frac{2}{x^2}$. At $x = 4.56$: $g'(4.56) = \frac{2}{20.7936} \approx 0.0962$. Since $|g'(4.56)| < 1$, the iteration converges near this root.

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13 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^2 - 3x - 1 = 0$ into the form $x = g(x)$ by making the linear term the subject.

From $3x = x^2 - 1$: $x = \frac{x^2 - 1}{3}$.

14 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^3 + x - 1 = 0$ into the form $x = \frac{1}{x^2 + 1}$, and verify algebraically that this is equivalent to the original equation.

From $x^3 + x - 1 = 0$: $x(x^2 + 1) = 1$, so $x = \frac{1}{x^2 + 1}$. Verifying: multiplying both sides by $(x^2 + 1)$ gives $x(x^2 + 1) = 1$, i.e. $x^3 + x = 1$, i.e. $x^3 + x - 1 = 0$, which matches the original equation exactly.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 11.01 Differential Equations

Use one iteration of the Newton-Raphson method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 4x - 2$, to four decimal places.

$f'(x) = 3x^2 - 4$. $f(2) = 8 - 8 - 2 = -2$. $f'(2) = 12 - 4 = 8$. $x_1 = 2 - \frac{-2}{8} = 2.25$.

16 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 12.01 Poisson Distribution

Continuing from $x_1 = 2.25$ in the previous problem, find x_2, to four decimal places.

$f(2.25) = 11.390625 - 9 - 2 = 0.390625$. $f'(2.25) = 3(5.0625) - 4 = 11.1875$.
$x_2 = 2.25 - \frac{0.390625}{11.1875} \approx 2.25 - 0.0349 \approx 2.2151$.

17 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 13.01 Continuous Random Variables

Use the Newton-Raphson method with $x_0 = 2.5$ to find x_1 for $f(x) = x^2 - 7$ (approximating $\sqrt{7}$), to four decimal places.

$f'(x) = 2x$. $f(2.5) = 6.25 - 7 = -0.75$. $f'(2.5) = 5$. $x_1 = 2.5 - \frac{-0.75}{5} = 2.5 + 0.15 = 2.65$.

18 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 14.01 Sampling & Estimation

Use the Newton-Raphson method with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 2x - 5$, to four decimal places.

$f'(x) = 3x^2 - 2$. $f(2) = 8 - 4 - 5 = -1$. $f'(2) = 12 - 2 = 10$. $x_1 = 2 - \frac{-1}{10} = 2.1$.

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19 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 15.01 Hypothesis Testing

Using $x_1 = 2.65$ from the previous problem for $f(x) = x^2 - 7$, perform a second iteration to find x_2, to four decimal places, and compare with $\sqrt{7} \approx 2.6458$.

$f(2.65) = 7.0225 - 7 = 0.0225$. $f'(2.65) = 5.3$. $x_2 = 2.65 - \frac{0.0225}{5.3} \approx 2.65 - 0.00425 \approx 2.6458$, matching $\sqrt{7}$ to four decimal places after just two iterations.

20 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 2.01 Exponentials & Logarithms

Explain why $x_0 = 1$ is an unsuitable starting point for the Newton-Raphson method applied to $f(x) = x^3 - 3x + 1$.

$f'(x) = 3x^2 - 3$. At $x_0 = 1$: $f'(1) = 3 - 3 = 0$. Since $f'(1) = 0$, the tangent line at $x_0 = 1$ is horizontal and never meets the x-axis, so the formula $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ involves division by zero — the method fails at this starting point.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Location of Roots & Sign Change

Show that the equation $x^3 - 2x - 5 = 0$ has a real root in the interval $[2, 3]$.

a. $f(2) = -1 < 0$ **and** $f(3) = 16 > 0$**, so by IVT a root exists ✓**

b. $f(2) = 1 > 0$ and $f(3) = -2 < 0$

c. $f'(x) > 0$ for all x

d. $f(2.5) = 0$

$f(2) = 8 - 4 - 5 = -1 < 0$. $f(3) = 27 - 6 - 5 = 16 > 0$. Since f is continuous on $[2, 3]$ and changes sign, the Intermediate Value Theorem guarantees at least one root.

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22 [Foundational] • Target: Location of Roots & Sign Change

Under which condition does a sign change of $f(x)$ between a and b fail to guarantee a root in (a, b)?

a. When $f(x)$ is not continuous on $[a, b]$ ✓

b. When $f'(x)$ changes sign

c. When $f(x)$ has an inflection point

d. When the interval is wider than 1 unit

If $f(x)$ has a vertical asymptote or jump discontinuity in (a, b) (e.g. $f(x) = 1/x$ between -1 and 1), a sign change occurs without any root.

23 [Foundational] • Target: Location of Roots & Sign Change

Use the sign change method to determine which interval of length 0.1 contains the root of $x^3 = 20$.

a. $[2.7, 2.8]$ ✓

b. $[2.5, 2.6]$

c. $[2.6, 2.7]$

d. $[2.8, 2.9]$

$2.7^3 = 19.683 < 20$. $2.8^3 = 21.952 > 20$. The root lies in $[2.7, 2.8]$.

24 [Foundational] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$

Rearrange the equation $x^3 - x - 1 = 0$ into an iterative formula $x_{n+1} = g(x_n)$.

a. $x_{n+1} = \sqrt[3]{x_n + 1}$ ✓

b. $x_{n+1} = x_n^3 - 1$

c. $x_{n+1} = \frac{1}{x_n^2 - 1}$

d. $x_{n+1} = x_n^3 + x_n$

$x^3 = x + 1 \Rightarrow x = (x + 1)^{1/3}$. Iterative formula is $x_{n+1} = \sqrt[3]{x_n + 1}$.

25 [Foundational] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$

State the condition on $g'(x)$ for the iteration $x_{n+1} = g(x_n)$ to converge to a root α.

a. $|g'(\alpha)| < 1$ ✓

b. $|g'(\alpha)| > 1$

c. $g'(\alpha) = 0$

d. $g'(\alpha) > 0$

Fixed-point iteration converges if and only if $|g'(x)| < 1$ in a neighborhood of the root α.

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26 [Foundational] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$

Starting with $x_1 = 1.3$, use $x_{n+1} = \sqrt[3]{x_n + 1}$ to calculate x_2 and x_3 correct to 3 decimal places.

a. $x_2 = 1.320, x_3 = 1.324$ ✓

b. $x_2 = 1.300, x_3 = 1.310$

c. $x_2 = 1.350, x_3 = 1.380$

d. $x_2 = 1.315, x_3 = 1.330$

$x_2 = (2.3)^{1/3} \approx 1.3200$. $x_3 = (2.3200)^{1/3} \approx 1.3238 \approx 1.324$.

27 [Foundational] • Target: Cobweb and Staircase Diagrams

If $0 < g'(\alpha) < 1$ at the root, what pattern is produced on the iteration diagram?

a. **A staircase diagram converging to the root** ✓

b. A cobweb diagram converging to the root

c. A diverging staircase diagram

d. An oscillating cycle

When the gradient $g'(x)$ is positive and less than 1, successive iterations approach the root monotonically from one side, forming a staircase diagram.

28 [Foundational] • Target: Cobweb and Staircase Diagrams

If $-1 < g'(\alpha) < 0$ at the root, what pattern is produced on the iteration diagram?

a. **A cobweb diagram spiraling inwards (converging)** ✓

b. A staircase diagram spiraling outwards

c. A cobweb diagram spiraling outwards (diverging)

d. A vertical asymptote

A negative derivative between -1 and 0 causes the iterates to alternate sides around the root while decreasing in distance, forming an inward-spiraling cobweb.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Cobweb and Staircase Diagrams

What happens when $|g'(\alpha)| > 1$?

a. **The iteration diverges away from the root** ✓

b. The iteration converges very quickly

c. The iteration oscillates between two fixed values

d. The root becomes imaginary

If $|g'(\alpha)| > 1$, the fixed point is repulsive (unstable), and the sequence diverges away from α.

30 [Intermediate] • Target: Newton-Raphson & Advanced Applications

State the Newton-Raphson iteration formula for finding roots of $f(x) = 0$.

a. $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ ✓

b. $x_{n+1} = x_n + \frac{f(x_n)}{f'(x_n)}$

c. $x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$

d. $x_{n+1} = \frac{x_n f'(x_n) - f(x_n)}{2}$

The Newton-Raphson formula is $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$.

31 [Intermediate] • Target: Newton-Raphson & Advanced Applications

Use one iteration of Newton-Raphson with $x_0 = 2$ to approximate the root of $f(x) = x^2 - 5$.

a. $x_1 = 2.25$ ✓

b. $x_1 = 2.20$

c. $x_1 = 2.30$

d. $x_1 = 2.15$

$f(2) = 4 - 5 = -1$. $f'(x) = 2x \Rightarrow f'(2) = 4$. $x_1 = 2 - \frac{-1}{4} = 2 + 0.25 = 2.25$.

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32 [Intermediate] • Target: Newton-Raphson & Advanced Applications

Under which circumstance will the Newton-Raphson method fail completely at an iteration step x_n?

a. When $f'(x_n) = 0$ (tangent is horizontal) ✓

b. When $f(x_n) = 0$

c. When $f''(x_n) = 0$

d. When x_n is negative

If $f'(x_n) = 0$, the tangent line is horizontal and never intersects the x-axis, causing division by zero in the formula.

33 [Intermediate] • Target: Newton-Raphson & Advanced Applications

The function f is defined for all $x \in \mathbb{R}$ by $f(x) = x^3 - 2x^2 - kx + 12$, where k is a real constant. The Newton-Raphson method is used to find an approximation to a root of the equation $f(x) = 0$. Given that an initial approximation $x_0 = 2$ produces a first approximation $x_1 = 4$, find the value of the constant k and the exact value of the second approximation x_2.

a. $k = 5, \quad x_2 = \frac{28}{9}$ ✓

b. $k = 5, \quad x_2 = \frac{44}{9}$

c. $k = 2, \quad x_2 = \frac{14}{5}$

d. $k = 7, \quad x_2 = \frac{84}{25}$

We solve the problem in two steps: **Step 1: Find the value of k** The Newton-Raphson iteration formula is: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ Differentiating $f(x) = x^3 - 2x^2 - kx + 12$ with respect to x: $f'(x) = 3x^2 - 4x - k$ Evaluating $f(x)$ and $f'(x)$ at $x_0 = 2$: $f(2) = 2^3 - 2(2^2) - 2k + 12 = 8 - 8 - 2k + 12 = 12 - 2k$ $f'(2) = 3(2^2) - 4(2) - k = 12 - 8 - k = 4 - k$ Using the iteration formula for x_1: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow 4 = 2 - \frac{12-2k}{4-k}$ Rearran.

34 [Intermediate] • Target: Locating roots by a change of sign

Using the Newton-Raphson method with $f(x) = x^3 - 53$ and $x_0 = -8$, find x_1.

a. $-\frac{779}{192}$

b. $\frac{971}{192}$

c. $-\frac{971}{192}$ ✓

d. -8

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{2x_0^3 + 53}{3x_0^2} = -\frac{971}{192}$.

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35 [Intermediate] • Target: Interval bisection

Using the Newton-Raphson method with $f(x) = x^3 - 48$ and $x_0 = 7$, find x_1.

a. 7

b. $-\frac{734}{147}$

c. $\frac{881}{147}$

d. $\frac{734}{147}$ ✓

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{2x_0^3 + 48}{3x_0^2} = \frac{734}{147}$.

36 [Intermediate] • Target: Fixed-point iteration

Using the Newton-Raphson method with $f(x) = x^3 - 10$ and $x_0 = 3$, find x_1.

a. $\frac{91}{27}$

b. 3

c. $-\frac{64}{27}$

d. $\frac{64}{27}$ ✓

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{2x_0^3 + 10}{3x_0^2} = \frac{64}{27}$.

37 [Intermediate] • Target: Convergence and divergence of iterations

Using the Newton-Raphson method with $f(x) = x^3 - 33$ and $x_0 = -3$, find x_1.

a. $-\frac{7}{9}$ ✓

b. $\frac{7}{9}$

c. $\frac{2}{9}$

d. -3

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{2x_0^3 + 33}{3x_0^2} = -\frac{7}{9}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rearranging equations for iteration • Cross-Lesson: 14.01 Sampling & Estimation

Using the Newton-Raphson method with $f(x) = x^3 - 2$ and $x_0 = 4$, find x_1.

a. $\frac{65}{24}$ ✓

b. $-\frac{65}{24}$

c. 4

d. $\frac{89}{24}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{2x_0^3 + 2}{3x_0^2} = \frac{65}{24}$.

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39 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 15.01 Hypothesis Testing

Using the Newton-Raphson method with $f(x) = x^3 - 56$ and $x_0 = 1$, find x_1.

a. 1

b. $\frac{61}{3}$

c. $-\frac{58}{3}$

d. $\frac{58}{3}$ ✓

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{2x_0^3 + 56}{3x_0^2} = \frac{58}{3}$.

40 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 2.01 Exponentials & Logarithms

Using the Newton-Raphson method with $f(x) = x^2 - 18$ and $x_0 = 5$, find x_1.

a. 5

b. $\frac{43}{10}$ ✓

c. $-\frac{43}{10}$

d. $\frac{53}{10}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 18}{2x_0} = \frac{43}{10}$.

41 [Challenging] • Target: Error bounds from bisection • Cross-Lesson: 3.01 Advanced Trigonometry

Using the Newton-Raphson method with $f(x) = x^2 - 14$ and $x_0 = 5$, find x_1.

a. $\frac{49}{10}$

b. $-\frac{39}{10}$

c. 5

d. $\frac{39}{10}$ ✓

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 14}{2x_0} = \frac{39}{10}$.

42 [Challenging] • Target: A well-designed word problem: designing a support strut • Cross-Lesson: 4.01 Differentiation

Using the Newton-Raphson method with $f(x) = x^2 - 13$ and $x_0 = 6$, find x_1.

a. 6

b. $\frac{61}{12}$

c. $\frac{49}{12}$ ✓

d. $-\frac{49}{12}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 13}{2x_0} = \frac{49}{12}$.

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43 [Challenging] • Target: Location of Roots & Sign Change • Cross-Lesson: 5.01 Integration

Using the Newton-Raphson method with $f(x) = x^2 - 22$ and $x_0 = -3$, find x_1.

a. $-\frac{25}{6}$

b. $-\frac{31}{6}$ ✓

c. $\frac{31}{6}$

d. -3

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 22}{2x_0} = -\frac{31}{6}$.

44 [Challenging] • Target: Fixed-Point Iteration $x_{n+1} = g(x_n)$ • Cross-Lesson: 6.01 Sequences & Series

Using the Newton-Raphson method with $f(x) = x^2 - 13$ and $x_0 = 2$, find x_1.

a. 2

b. $-\frac{17}{4}$

c. $\frac{21}{4}$

d. $\frac{17}{4}$ ✓

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 13}{2x_0} = \frac{17}{4}$.

45 [Challenging] • Target: Cobweb and Staircase Diagrams • Cross-Lesson: 7.01 Vectors

Using the Newton-Raphson method with $f(x) = x^2 - 18$ and $x_0 = 4$, find x_1.

a. 4

b. $-\frac{17}{4}$

c. $\frac{21}{4}$

d. $\frac{17}{4}$ ✓

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 18}{2x_0} = \frac{17}{4}$.

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